The Wingmen

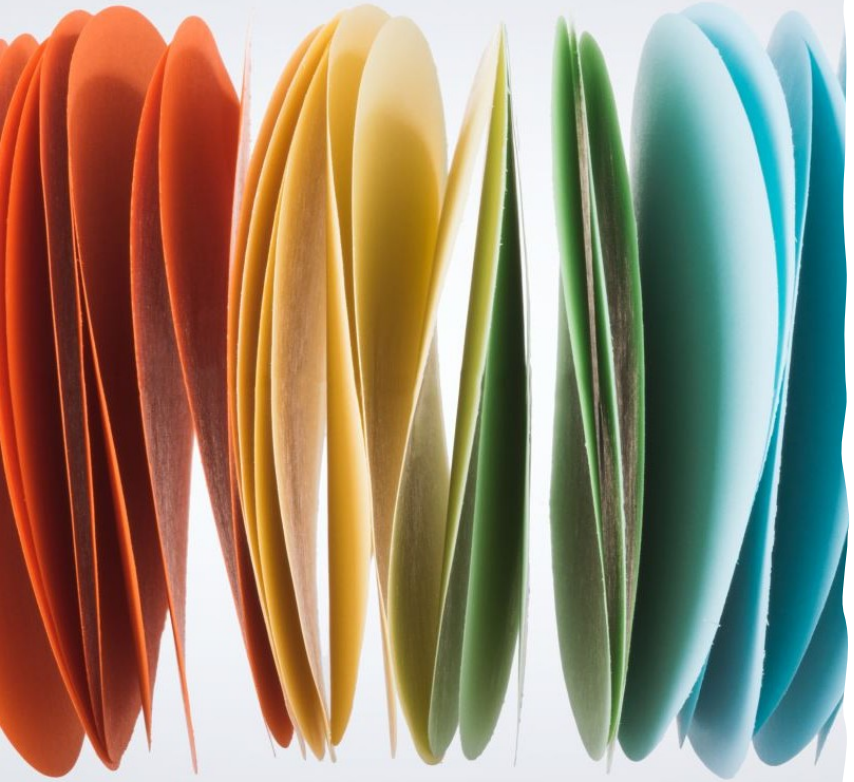
- Group Members:
- Collin Atkins
- Preston Bugh
- Blake Miller
- Desmond Rubio





Glider Constraints and Requirements

- Fuselage length must be greater than 0.5 m and less than 1.5 m
- Wingspan must be greater than 0.5 m and less than 1.5 m
- We < 15 N
- No thrust devices
- No rotation of the glider except for natural (but undesired) turning from the intended straight flight path due to imbalance
- Hand launch from stationary position; top of release point no higher than 2.5 m
- Must fly under an arch barrier (5 m wide by 3 m high) placed 5 m downstream from the launch point



Glider Constraints and Requirements

- Price (purchased or built) must be less than $50
- Acceptable materials: balsa wood (and other light woods), cardboard, poster board, paper products, light plastics and foams, thin carbon-fiber or composite rods, foil, monokote, plastic wrap, tape, glue, wire, string/twine, rubber bands, simple binders/clips
- Items not allowed: items not listed above including but not limited to metal components (other than thin wire, coins or small balance weights), glass, ceramics (or brittle materials that can break or shatter), sharp items, large pieces of composite materials, liquids, combustible items, and energy devices (e.g., batteries)

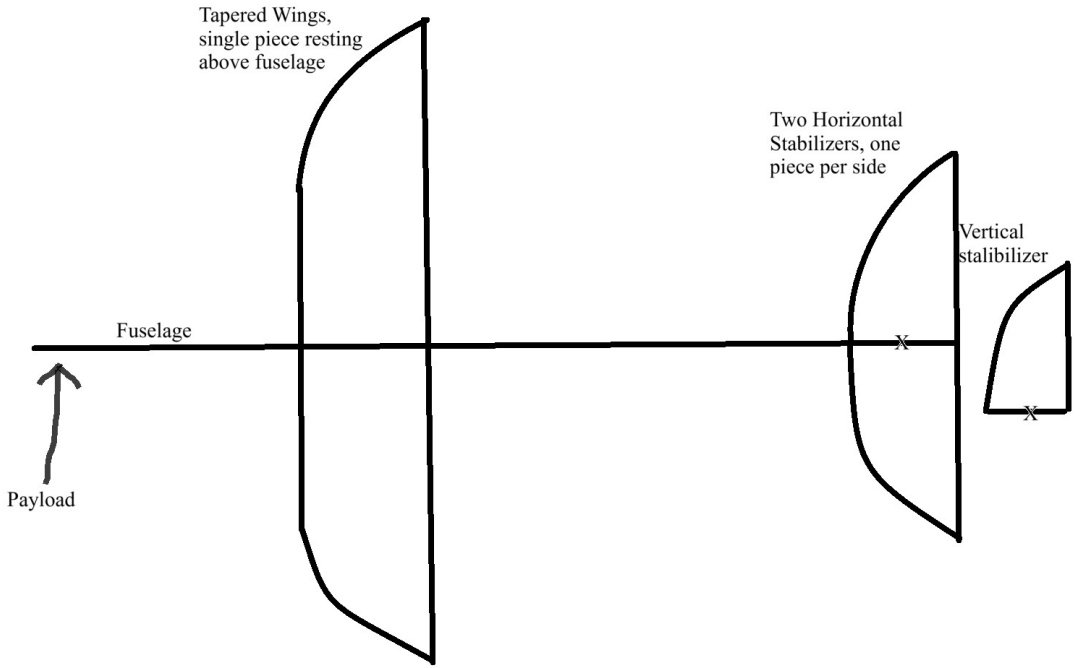


Design Philosophy

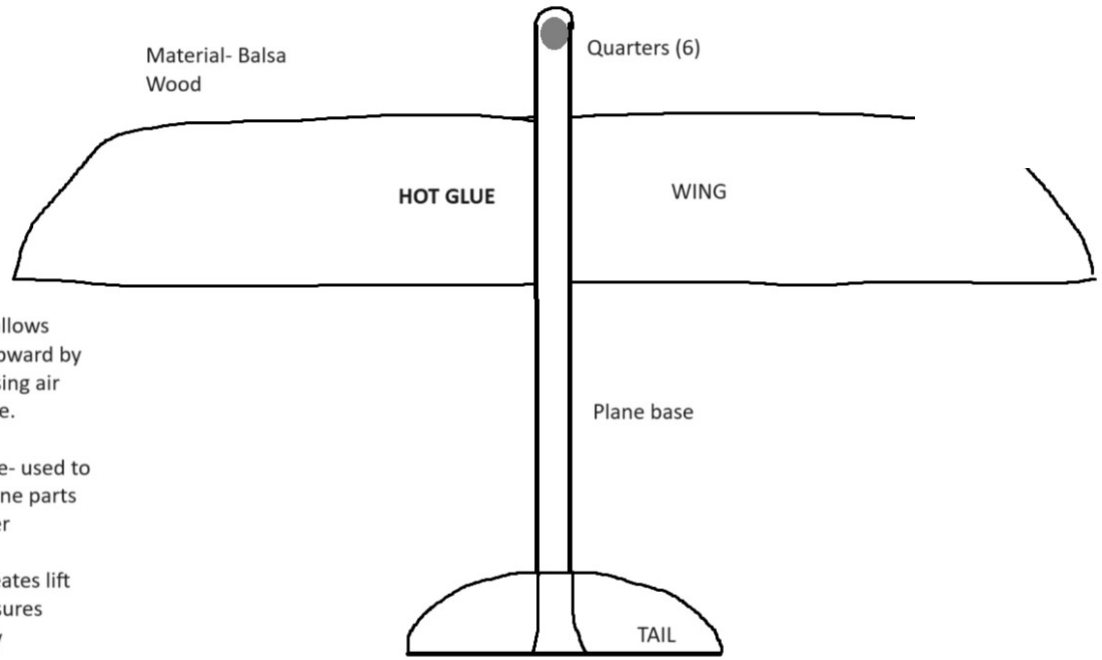
- The main area of focus for the glider is to maintain height in accordance with a specific trajectory.
- Our glider was built rather than purchased because putting the glider together piece by piece allows us to better adjust and modify the glider for its objective.
- Resources used: tape, glue, balsa wood, quarters.
- Group meetings:
  - Wednesday, November 15th, 5:15 PM, at Rodgers Library (in-person work layout discussion).
  - Monday, November 27th, 4:30 PM, at Ridgecrest (glider testing).
  - Late October-December 1st, Snapchat group chat (group communication and in-person meeting scheduling).

Design Philosophy

Prebuild Sketch



PREBUILD SKETCH



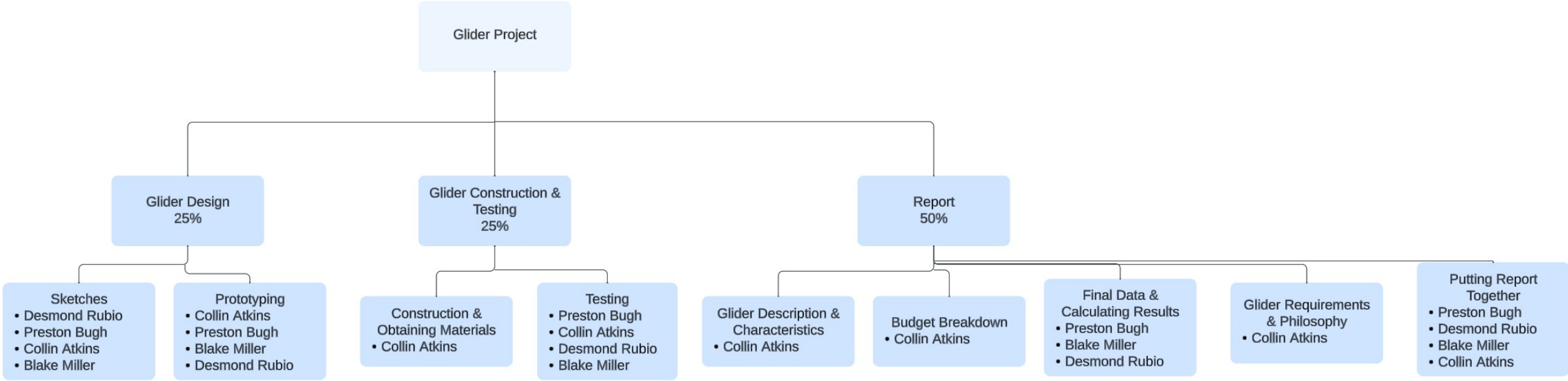
Wing- allows flight upward by decreasing air pressure.

Hot glue- used to seal plane parts together

Tail- creates lift and ensures stability

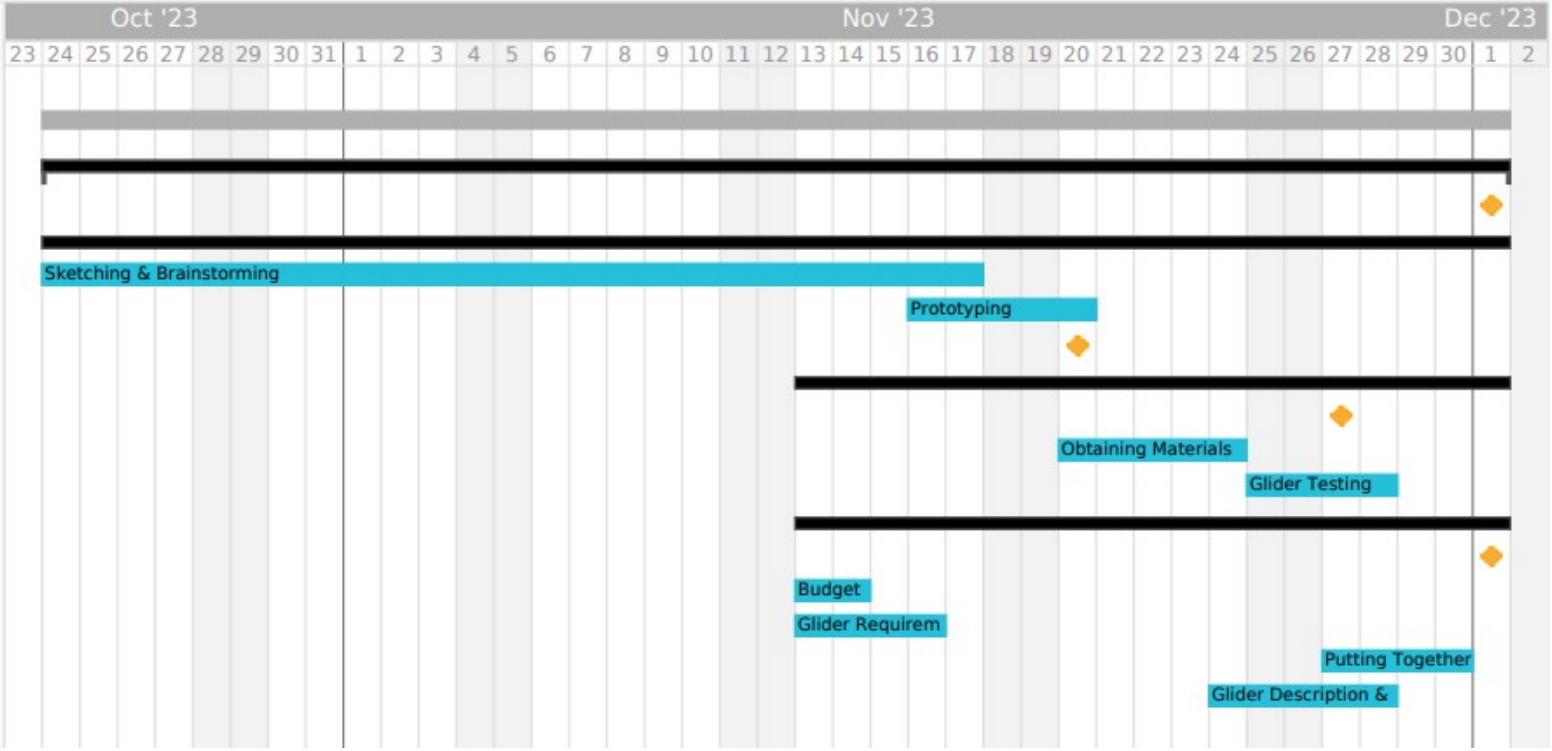
Plan Base- frame of the aircraft

Work Breakout Structure

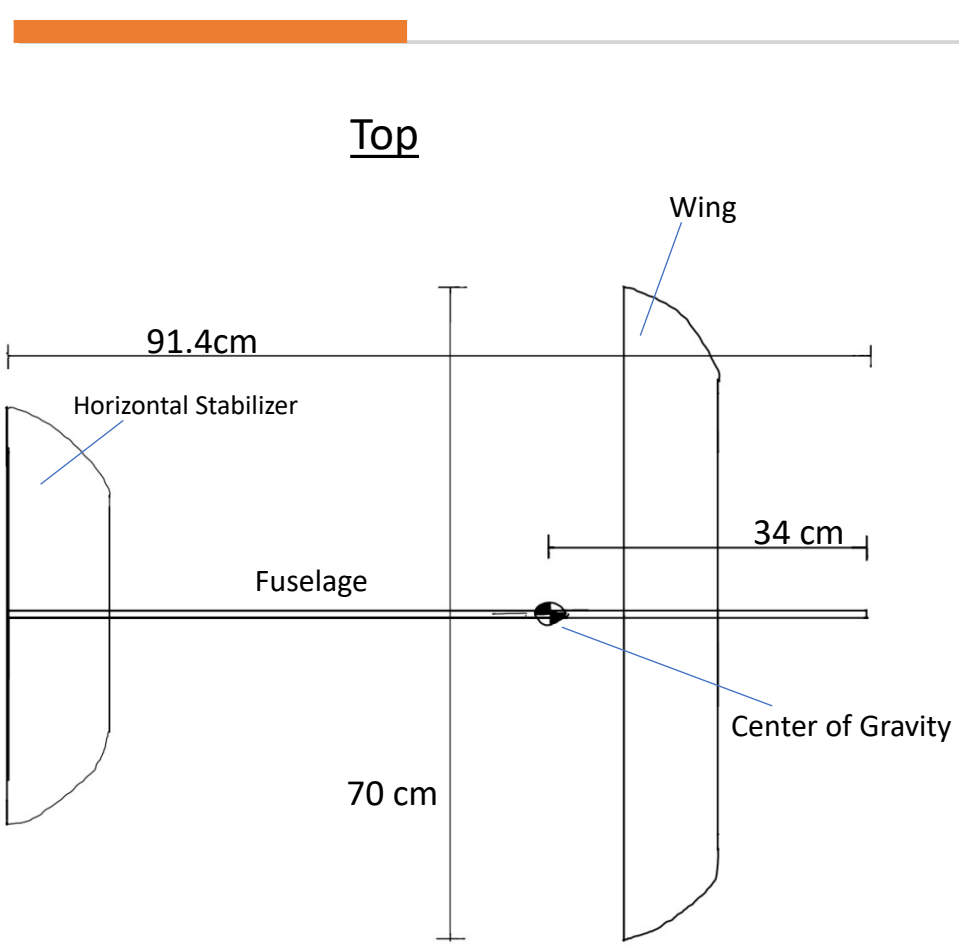


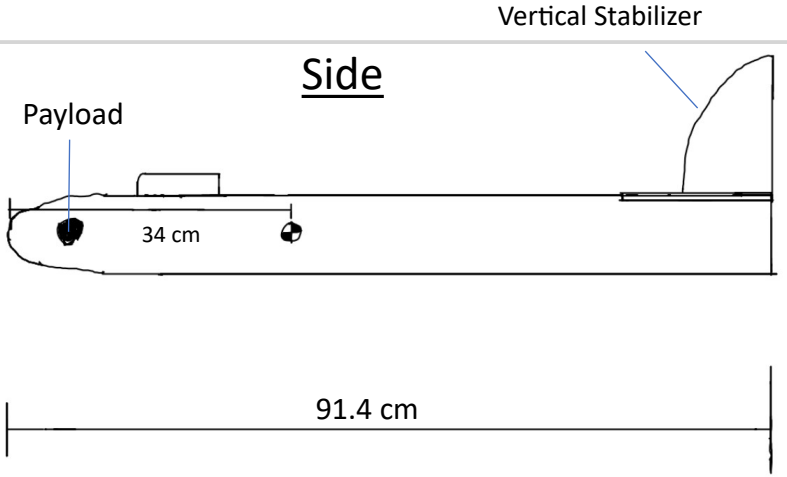
Gantt Chart

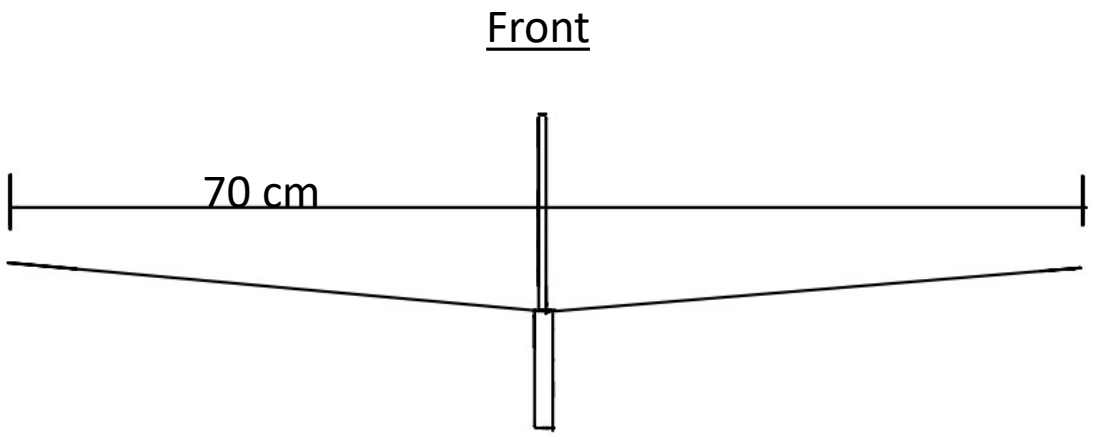
Gantt Chart	start	end	0h	100%
Glider Project	**10/24/23**	**12/01/23**	**0h**	**100%**
Project Finished	12/01	12/01	0	100%
Glider Design	**10/24/23**	**12/01/23**	**0h**	**100%**
Sketching & Brainstorming	10/24	11/17	0	100%
Prototyping	11/16	11/20	0	100%
Glider Designed and Ready to Build	11/20	11/20	0	100%
Glider Construction & Testing	**11/13/23**	**12/01/23**	**0h**	**100%**
Glider Fully Constructed and Tri...	11/27	11/27	0	100%
Obtaining Materials & Constructi...	11/20	11/24	0	100%
Glider Testing	11/25	11/28	0	100%
Report	**11/13/23**	**12/01/23**	**0h**	**100%**
Report Finished	12/01	12/01	0	100%
Budget Breakdown	11/13	11/14	0	100%
Glider Requirements & Philoso...	11/13	11/16	0	100%
Putting Together & Calculating...	11/27	11/30	0	100%
Glider Description & Characteri...	11/24	11/28	0	100%



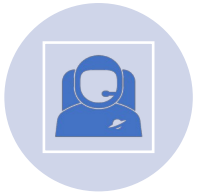
Post-Build Sketches







Description of the Glider



Our glider was designed and built instead of purchased to allow us to implement our ideas onto the glider and modify it easier. It's also cheaper.



The payload of quarters was mounted onto the curved nose of the glider using tape with an equal number of quarters on each side touching each other to balance out the weight to avoid glider rotation.



The primary material used was balsa wood because it's easy to manipulate and very light.



Each side of the glider was cut and assembled equally to ensure it did not lean to the left or right more than the other and the effect from the weight of the quarters on the glider was carefully checked to make sure the center of gravity was in the middle of the glider.



The glider was assembled by using posterboard as a template to cut the balsa wood and using glue to attach the balsa wood pieces together and tape to attach the quarters and glider together.

Glider Characteristics

1		Glider Characterisitcs
2	Mass (empty) [g]	72.0 g
3	CG location relative to wing LE [cm], empty	21.2 cm
4	CG location relative to wing LE [cm], with payload	10.0 cm
5	Wingspan [cm]	70.0 cm
6	Wing area [cm2]	579.0 cm^2
7	Wing aspect ratio	8.46
8	Wing average chord [cm]	8.27 cm
9	Wing taper ratio	0
10	Wing leading edge sweep [°]	90°
11	Distance from CG to tail LE [cm]	48.9 cm
12	Horizontal tail area [cm2]	261.3 cm^2
13	Vertical tail area [cm2]	130.65 cm^2

Flight Data

Test Day Data Sheet
(provided day of test)

Team Name	The Wingmen		
Team Members	Preston Bugh, Blake Miller, Desmond Rubio, Collin Atkins		
Test Date	11/28/2023		
Trial	1	2	3
Estimated Wind Speed [mph]	5	5	N/A
Pressure [in-Hg]	30.37	30.37	N/A
Temperature [F]	38	38	N/A
Mass of Glider, empty [g]	72	72	N/A
Laungh Height [in]	70	70	N/A
Range [ft]	26	30	N/A
Flight Time [s]	2.5	1.5	N/A
Passengers	4	4	N/A

Table of Calculated Results

Estimated Wind	Temperature	Pressure	Glide Speed	Lift Coefficient	Glide Angle	L/D	Number of Quarters Carried	Wing Loading with Payload	Score
2.24 m/s	277 K	103 kPa	6.1 m/s	$7.40*10^{-3}$	11°	5.14	4	15.9 N/m^2	0.51 m/N

Budget Breakdown

- Budget=$50
- Actual amount spent=$23.74
  - 2 1/8 x 4 x 36 in balsa wood boards = $9.98 + taxes ($0.6986) = $10.6786 ≈ $10.68
  - 1 3/16 x 1 x 36 in balsa wood board = $3.59 + taxes ($0.2513) = $3.8413 ≈ $3.84
  - 1 1/8 x 2 x 36 in balsa wood board = $2.84 + taxes ($0.1988) = $3.0388 ≈ $3.04
  - 1 superglue tube = $2.49 + taxes ($0.1743) = $2.6643 ≈ $2.66
  - 1 duct tape roll = $3.29 + taxes ($0.2303) = $3.5203 ≈ $3.52
  - 4 quarters = $1.00

